

Negative indices



- 1
- a i 10 ii -7 b i 2 ii -4 c i 13 ii -10 d i -5 ii -8

2

a

2^5	2^4	2^3	2^2	2^1	2^0	2^{-1}
32	16	8	4	2	1	$\frac{1}{2}$

b

10^5	10^4	10^3	10^2	10^1	10^0	10^{-1}
100 000	10 000	1000	100	10	1	$\frac{1}{10}$

c

3^3	3^2	3^1	3^0	3^{-1}	3^{-2}	3^{-3}
27	9	3	1	$\frac{1}{3}$	$\frac{1}{9}$	$\frac{1}{27}$

- 3
- a $\frac{1}{6^{10}}$ b $\frac{1}{73^{14}}$ c $\frac{1}{(-9)^7}$ d $\frac{1}{9}$
- e $\frac{1}{(17)^6}$ f $\frac{1}{55}$ g $\frac{1}{(-12)^8}$ h $-\frac{1}{45^5}$
- i $-\frac{1}{8^{11}}$ j $\frac{1}{(-20)^3}$ k $\frac{1}{7^6}$ l $\frac{1}{-5}$

- 4
- a 3^{-1} b 37^{-1} c 5^{-1} d 4^{-7}
- e -15^{-3} f 10^{-5} g $(-24)^{-10}$ h 25^{-3}
- i 13^{-11} j 7^{-8} k 16^{-12} l $(-45)^{-7}$

- 5
- a -2 b -5 c -3 d -3
- e -11 f -3 g -4 h -5

6

a 5^{14}

b 7^{-12}

c 7^{-7}

d 5^5

e 9^{-12}

f 5^{22}

g 10^{-1}

h 100^{-1}

i 2^{-2}

j 5^{-16}

k 19^{18}

l -8^{24}



7

a $\frac{13}{9}$

b 16

c $\frac{19}{18}$

d $\frac{81}{4}$

8

a 0

b $0^{-4} = \frac{1}{0^4} = \frac{1}{0}$ and dividing by zero is impossible.

9

a 1

b 16

c 27

d $\frac{1}{25}$

e $\frac{1}{4096}$

f $\frac{1}{144}$

g $\frac{1}{225}$

h $\frac{1}{256}$

i $\frac{1}{49}$

j $\frac{1}{9}$

k 125

l $\frac{1}{-7}$